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Senario Based Test

1. Cloud Based Big Data Architecture for a Retail Company

1. Understanding the Scenario:

A retail company collects terabytes of customer transaction data every day from online and offline stores. The company wants to analyze this data in near real time to understand purchasing patterns, recommend products, and manage inventory efficiently.

2. Proposed Cloud Architecture:

The company can use a cloud-based big data architecture consisting of the following components:

1. **Data Sources**: Collect data from online shopping websites, physical stores, billing systems, and customer transactions.
2. **Data Ingestion**: Use Apache Kafka or a cloud messaging service to collect transaction data continuously.
3. **Distributed Storage**: Store raw and processed data in a cloud data lake using Amazon S3, Azure Data Lake Storage, or Google Cloud Storage.
4. **Distributed Processing**: Use Apache Spark to process large datasets across multiple computers simultaneously.
5. **Analytics and Recommendations**: Analyze customer purchasing patterns and identify frequently purchased products.
6. **Data Warehouse**: Store cleaned and structured data in Amazon Redshift, Google BigQuery, or Azure Synapse Analytics for reporting.
7. **Visualization**: Use Power BI or Tableau to display sales trends, inventory levels, and customer behavior.

3. Integration Of Batch and Stream Processing

Batch processing processes large amounts of stored data at scheduled intervals. For example, the company can analyze the previous months sales to identify seasonal purchasing patterns.

Stream processing analyzes incoming data continuously. For example, when a product sells rapidly, the system can update its sales statistics and inventory information almost immediately.

Both methods can be integrated by storing incoming data in a data lake. Stream processing handles immediate analysis, while batch processing analyzes historical data. Their results can be combined to provide accurate business insights.

4. Use Of Data Lakes and Data Warehouses.

A data lake stores large amounts of raw data in different formats, including structured, semi-structured, and unstructured data. It is suitable for storing transaction records, clickstream data, and customer activity logs. A data warehouse stores cleaned, structured and organized data for reporting and business analysis.

The company can store all collected data in a data lake, process and clean it, and transfer the required information into a data warehouse. This approach supports both advanced analytics and business reporting.

5. Scalability and Cost Efficiency.

The company should use the following techniques:

Auto-Scaling: Automatically increases or decreases computing resources according to workload

Distributed processing: Processes large datasets using multiple computers

Data compression: Reduces storage requirements.

Life cycle Policies: Move older, less frequently accessed data to low cost storage.

Serverless analytics: Avoids maintaining dedicated servers for certain processing tasks.

Resource monitoring: Identifies unused resources and reduces unnecessary expenses

Conclusion:

A cloud-based big data architecture using distributed Storage, Apache Spark, batch processing, and stream processing, and stream processing enables the retail company to analyze large volumes of data efficiently. Combining data lakes and data warehouses improves decision-making, product recommendations, inventory management, Scalability, and cost efficiency.

2. Secure Cloud Architecture for Healthcare Data.

1. Understanding the Scenario:

A healthcare organization wants to migrate patient records and medical imaging data to the cloud. It must protect sensitive medical information, comply with privacy regulations, and allow authorized doctors to access records Securely from different locations.

2. Proposed Secure Cloud Architecture

The organization should implement a Secure cloud architecture with encryption, identity and access management, centralized monitoring, and controlled access to patient records.

3. Encryption, IAM, and Key Management.

Encryption: Encryption protects Sensitive data from unauthorized access.

Encryption at rest: Encrypts patient records and medical images Stored in cloud databases and storage.

Encryption in transit: Uses TLS to protect data transmitted between doctors, applications, and cloud Services.

IAM: IAM controls who can access cloud resources.

Assign individual accounts to doctors and healthcare Staff.

Apply role-based access Control (RBAC)

Give each user only the permissions required for their job

Key Management: A cloud key management Service Stores and manages encryption keys Securely. Access to keys should be restricted, and key rotation should be performed according to Security policies.

4. Auditing, Logging, and Compliance Monitoring

The organization should maintain detailed records of access to patient information.

Audit logs: Record who accessed, modified, or deleted patient records.

Security monitoring: Detects unusual activities, suspicious logins, and repeated failed authentication attempts.

Compliance monitoring: Checks whether security configurations follow applicable healthcare privacy regulations.

Alerts: Notifys administrators about unauthorized access attempts or potential data breaches

Regular audits: Review permissions, access records, and security policies.

5. Preventing Unauthorized Access and Data Breaches.

Use strong passwords and multi-factor authentification

Apply least-privilege access and role-based permissions

Encrypt patient records & medical images

Conduct Vulnerability assessments and security audits

Maintain encrypted backups and test recovery procedures

Conclusion:

A secure cloud architecture using encryption, IAM, Keys management, audit logging, and continuous monitoring protects confidential healthcare informations.

3. Online Ticket Booking - Serverless Computing.

1. Serverless architecture:

Use an API Gateway, Serverless functions, a managed database, and cloud storage to build the ticket booking application.

2. Event-driven Computing:

Serverless functions automatically execute when booking events occurs. Automatic scaling handles sudden-traffic increase.

3. Managed Services:
API Gateway manages requests, managed databases store booking details, and message queues handle bookings

4. Advantages and limitations:
Advantages include automatic scaling, reduced server management, and pay-per-use pricing. Limitations include cold starts, execution limits, and dependence on the cloud provider.

4. Multinational Company - ETL/ELT pipeline

1. Cloud-based ETL/ELT pipeline:
Collect data from sales, finance, manufacturing, and customer support systems. Extract, transform, and load the data into a centralized cloud data warehouse.

2. Data quality and governance:
Validate data, remove duplicates, maintain metadata, and apply data governance policies to ensure accuracy, consistency, and security.

3. Workflow Orchestration:
Use workflow orchestration tools to automate, schedule, and monitor data integration tasks.

4. Query and Storage Optimization:
Use indexing, partitioning, compression, and efficient storage tiers to improve query performance and reduce costs.

5. Logistics Company - Machine Learning Analytics.

1. Cloud-based ML architecture:
Collect historical shipment data, weather information, and live GPS data. Store and process the data in the cloud, train ML models, and predict delivery delays.

2. Big data technologies:
Use distributed processing for large-scale model training, and deploy trained models to provide real-time.

3. Cloud - native AI/ML Services:

Use managed ML Services for model training, deployment, monitoring, and performance evaluation.

4. Data Ingestion and model Updates.

Use Streaming tools for live GPS data, feature engineering to prepare useful inputs, and continuous model retraining to maintain prediction accuracy.